



**National Federated
Compute Services**
NetworkPlus

Isambard Summit
24-25 March 2026

Federated AI Container Registry Feasibility Study

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Cambridge**

UCL



BriCS





**National Federated
Compute Services**
NetworkPlus

NFCS Overview

What the main project is about

The National Federated Compute Services NetworkPlus initiative is engaging compute service users and providers who are focused on the three pillars of governance, technology, and community, to develop a roadmap for a national federation of compute services to support the UK's digital research infrastructure.





Projects

1 Inclusive Futures: User stories and mapping pathways for National Federated Compute Services	2 Federated AI application container platform / registry feasibility study	3 Enhancing HPC adoption through user-centred design
4 Federating Everything, Everywhere, All at once	5 UKRI research data landscape survey	6 Bridging the Gap: Aligning project administration with access to digital research infrastructure
7 Towards a common data infrastructure for laboratory science	8 Federated Edge-HPC architectures for AI workflows in privacy- sensitive and real- time domains	9 UNITED: A Framework for federated computing roadmaps
10 Exploring the requirements and technologies for a data centre API	11 Federation of compute and infrastructures in the arts and humanities	12 A federated bespoke AI-assisted helpdesk for DRI facilities
13 Surveying accessibility of federated compute for junior researchers	14 Job Orchestration using Constellation on Heterogeneous HPC resources	15 Federated IAM for existing infrastructures
16 Evaluation of Secure Federated Kubernetes Storage for Trusted Research Environments	17 Net-Zero and Circular Economy Federation: Evidence-based Policy Roadmap for Carbon-Aware Compute in the Built	18 Federated data movement
19 DRI Federation Cybersecurity Roadmap development	20 Identifying Barriers for Biology Researchers Using Federated HPC Services	21 Exploring the governance requirements for enabling UK DRIs to adopt MyAccessID
22 Supercomputers and Superpositions: Making Quantum Accelerators Accessible Within HPC Frameworks	23 Tokenised Authentication for Research Computing Services (TARCS)	24 FAIR-Compute: A Roadmap for Fair and Efficient Allocation of Federated Digital Research Infrastructure
25 ACCORD (A Community for Contract Regulation for Data)	26 Federated data access across the DRI	

- Two rounds of funding
- 13 Phase 1
- 13 Phase 2

Outcomes



Each Project produces their report (by 31 July 2026) published with a DOI

Evidence is synthesised into a main report



**National Federated
Compute Services**
NetworkPlus

AI Container Registry Feasibility Study

Our project



Project team

Project Leads

- **Chris Edsall**

Co-leads

- **Sylvie Ramos (UCL)**
- **Rich Gilham (BriCS)**

Project Team Members

- **Frederic Brochu-Williams**
- **Adam Wallford**
- **Deepak Aggarwal**
- **Ethan Williams**



Work Packages

Work Package	Goals
WP1	Community Engagement & Requirements Elicitation
WP2	Landscape Analysis of Existing Tools & Technologies
WP3	Technical Challenges
WP4	Governance Challenges
WP5	Synthesis and Roadmap Contribution
WP6	Proof-of-Concept Evaluation

Activities

- NFCS Autumn Conference

- Workshop 1 at HPC-SIG

- ✓ **58 attendees** (25 online)
Audience: **sysadmins, infrastructure & resource providers**
- ✓ Focus: **user needs, operational concerns, portability**

- Workshop 2 at UCL (TRE use cases)

- ✓ **29 attendees** (8 online)
Audience: AI researchers working on TREs
- ✓ Focus: containers in TREs, real-world constraints

- NFCS Spring Conference

- Workshop 3 at Isambard-AI Days

- ✓ **12 attendees** (in-person only)
Audience: **sysadmins, infrastructure & resource providers**
- ✓ Focus: open discussion around authentication and authorization, container vs module share, hosting and governance

- NFCS Conference (6 and 7 July, London)

- Workshop 4 at Cambridge (21 July, Cambridge)

- ✓ Save the date!
- ✓ More info at <https://nfcs-networkplus-ai-containers.github.io/workshop4-website-github.io/index.html>

- NFCS Conference (10 and 11 September, Birmingham)



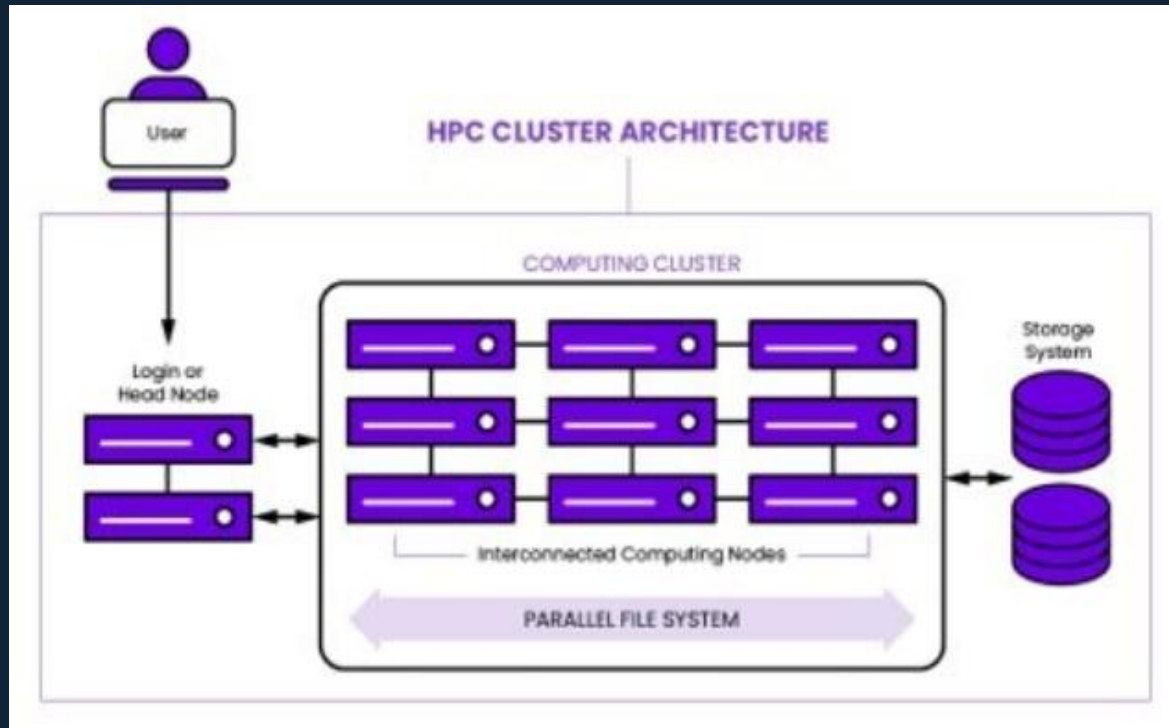


WP2 initial comparative studies

Comparison

Registry	SSO	Multi tenancy	Security scanning	Access control	GUI	Self hosted	Stores other artifacts	Open source	Free to use	Rate limits
GH Packages	Limited	Limited	No	Limited	Yes	No	Yes	No	Limited	
Gitlab	Yes	Limited	Yes	Limited	Yes	Optional	Yes	Yes	Yes	
Docker hub	Yes		Yes	Limited	Yes	No	No	No	Limited	
Distribution	Yes	No	No	Limited	No	Yes	No	Yes	Yes	
JFrog	Yes	Yes	Yes	Yes	Yes	Optional	Yes	No	Limited	
Harbor	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	
Nexus	Yes			Yes	Yes	Yes	Yes	Limited	Limited	
Pulp	Limited	No	No	Yes	No	Yes	Yes	Yes	Yes	
Quay	Yes	Yes	Yes	Yes	Yes	Optional	No	Yes	Yes	

WP3 Technical challenges

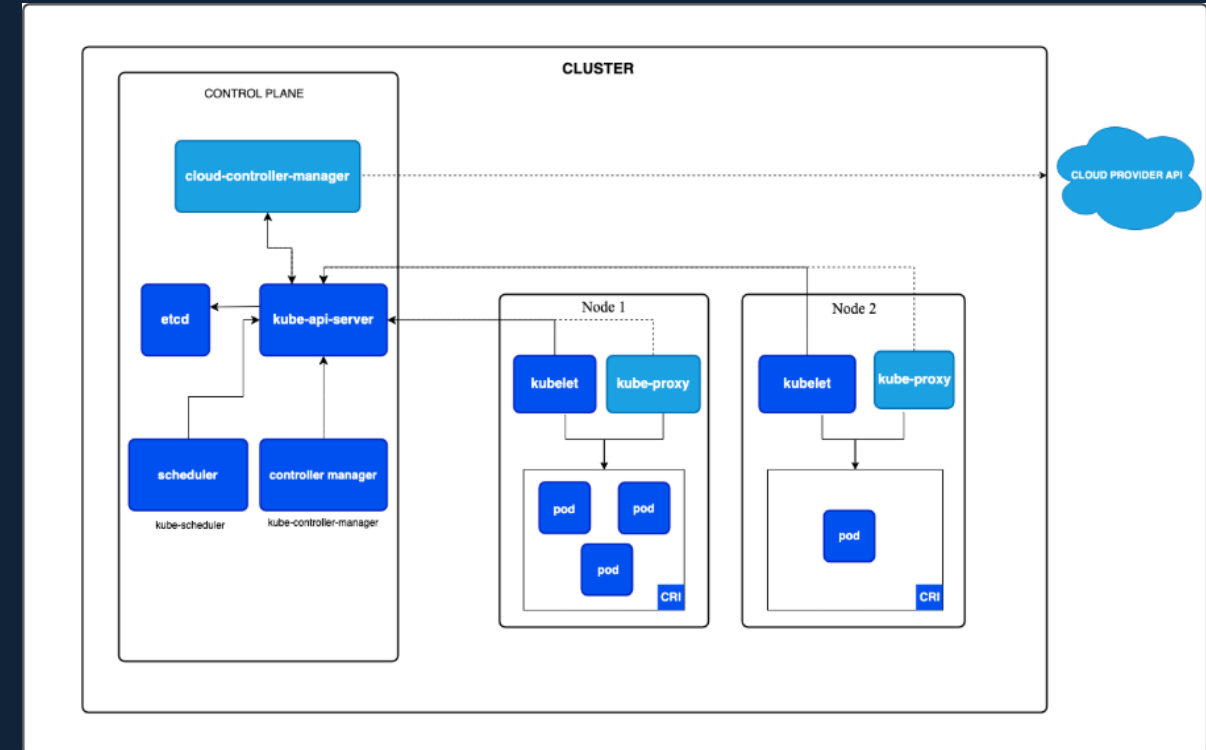


Source: <https://www.weka.io/learn/guide/hpc/what-are-hpc-and-hpc-clusters/>

Container engines: Apptainer, podman

Format: OCI vs SIF

Build tools: Spack, EasyBuild



Source: <https://kubernetes.io/docs/concepts/architecture/>

Container runtime: several available for example containerd, CRI-O ... or vendor specific (NVIDIA runtime)

Format: OCI

WP3 Technical challenges



Authentication/Authorization (MyAccessID)



Storage
network

UCL



Storage
network

Cambridge



Storage
network

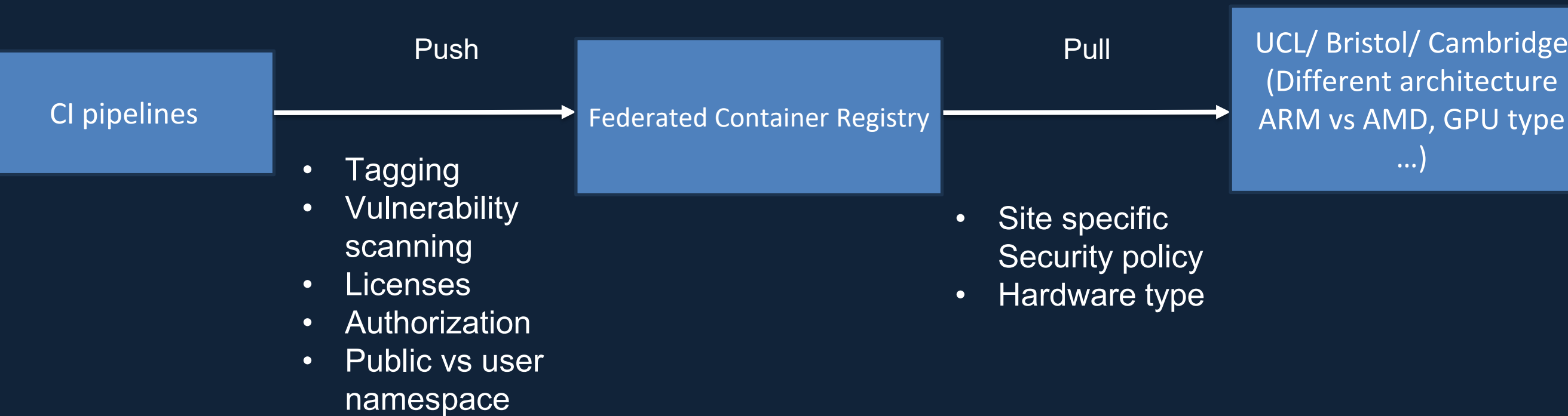
Bristol

Suggested:

- One container Registry per site
- Closer to the compute: Reduced pull time (large containers)
- Multi-site deployment (replication policy): more resilient (if issue with one site)
- Drawback: cost (infrastructure + operational)



WP4 Governance



Governance structure:

Chair

Package/Container manager

Representation from each site

Approved policies and user guidelines



Conclusion



Survey



**Workshop Registration
form**